

REAL ESTATE DATA ANALYTICS · SPRING 2026

The Anatomy of Neighborhood Stress

How ODIS Domain Scores Capitalize Into U.S. Housing Markets

Aryaman & Yelnur · April 2026

Research Question

Do different domains of neighborhood stress — economic, education, health, and crime — capitalize into U.S. housing markets at the same rate? And do they affect price levels differently from affordability?

Why this matters

- Most empirical work on neighborhood quality uses one indicator at a time — school test scores, crime, commute.
- ODIS gives us a multi-domain composite covering ~23,600 school neighborhoods, letting us decompose which stressors price in.
- Findings inform housing-policy targeting and how investors price neighborhood-quality risk.

Data at a glance

23,599

Public high-school
neighborhoods

17

Underlying stress
indicators

4

Domains used as
predictors
(econ, edu, health, crime)

3,167

Counties after
aggregation

Literature Review

Rent ↔ socioeconomic stressors

Ruivoa · Bryant Univ., Empirical Econ. Bulletin

Regression on U.S. cities: poverty rate, density, and supply each shape rent — establishes the regression-based precedent for our framework.

Data-driven migration & rent stress

Multiple authors · ResearchGate, Dec 2025

Spatial regression + ML show migration, stagnant incomes, and tight supply jointly amplify displacement — validates a composite stressor index over single variables.

Crime ↔ housing prices

Kallberg & Shimizu · J. Real Estate Finance & Econ., 2026

Seattle 2008–2020, 911 calls as IV. Crime → −0.55% on prices without spatial controls; +0.80% with them. Crime's effect depends on neighborhood wealth.

Crime → property values: lit survey

Meta-analysis · ResearchGate, 2017

Synthesizes hedonic literature: property-value elasticity to crime is −0.15 to −0.35; ZIPs in top decile of crime reduction saw 7–19% price gains in the 1990s.

Housing & health are intertwined

CBPP policy brief · June 2022

Moving to Opportunity: relocation to lower-poverty neighborhoods improved health outcomes. Medicaid expansion → fewer evictions of low-income renters.

Inflation pressures on renters

Harvard JCHS · April 2023

Census Household Pulse: ½ of renters and ⅔ of low-income renters report high financial stress. CPI rent peaked at 6% in Q4 2022 (and is structurally lagged).

Gap our study fills: prior work treats stressors in isolation. We unify economic, education, health, and crime under one composite framework, applied at county level.

What goes into each domain

ECONOMIC	EDUCATION	HEALTH	CRIME
• Unemployment (16+) • Child poverty (0–17) • Broadband access • Single-parent households	• Less-than-HS share • 2-yr-college-or-higher share • Linguistic isolation	• Healthcare access (kids) • Infant mortality • SNAP receipt • Low birth weight • Lead exposure risk	• Violent crime rate • Jail incarceration rate

Each domain = equal-weighted average of its components, rescaled 0–100. Housing domain (vacancy, affordability, parks) deliberately excluded — its 'affordability' indicator literally is rent burden.

Data pipeline & merge

ODIS v2

School-level
→ aggregated to
county
(median across
schools)

Zillow ZHVI

County-level
latest monthly value
(2026-03)

ACS 5-yr (2023)

Median rent, rent
burden,
income, pop, density,
vacancy, Gini

FHFA HPI

5-year cumulative
appreciation
(robustness)



Inner-join on 5-digit FIPS county code

Final analysis dataset

- n = 2,913 U.S. counties (complete-case across all four ODIS domains, ZHVI, ACS, controls)
- 254 counties dropped: 222 missing crime score (FBI UCR coverage gaps), 32 missing ZHVI (small rural counties + Puerto Rico)
- 49 states represented (DC included; PR excluded)

Formal hypotheses

Primary — Joint significance

$H_0: \beta_{\text{econ}} = \beta_{\text{edu}} = \beta_{\text{health}} = \beta_{\text{crime}} = 0$

H_1 : at least one $\beta \neq 0$

Test: Wald χ^2 on the four ODIS coefficients, jointly, in each fully-specified model.

Secondary — Cross-domain comparison

$H_{0_eq}: \beta_{\text{econ}} = \beta_{\text{edu}} = \beta_{\text{health}} = \beta_{\text{crime}}$ (equal capitalization)

H_{1_eq} : at least one pair differs

Tests whether the four domains move outcomes by the same magnitude

Run separately for each of four outcomes: log(ZHVI), log(median rent), rent burden share, FHFA 5-yr appreciation.

Empirical strategy

$$\log(Y_c) = \alpha_{\text{state}} + \beta_1 \cdot \text{econ}_c + \beta_2 \cdot \text{edu}_c + \beta_3 \cdot \text{health}_c + \beta_4 \cdot \text{crime}_c + \gamma' X_c + \varepsilon_c$$

Predictors & outcomes

Predictors (RHS):

- Four ODIS domain percentile ranks (0–100):
econ, edu, health, crime

Outcomes (4 separate regressions):

- log(ZHVI) — owner-occupied price level
- log(median rent) — rental price level
- Rent burden share — affordability
- FHFA 5-yr appreciation — price dynamics

Controls & inference

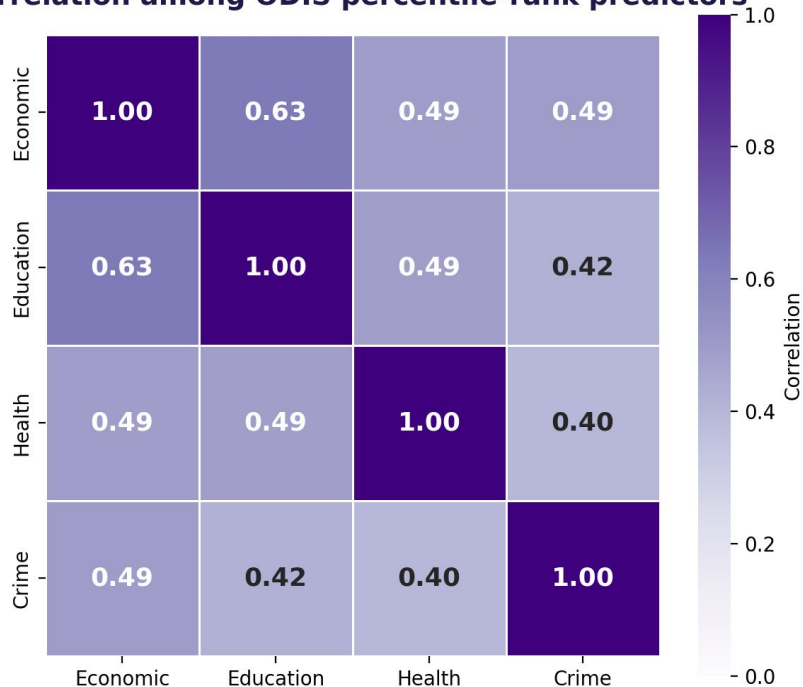
Controls (X_c):

- State fixed effects (49 dummies)
- log(population), log(median HH income)
- log(population density) ← addresses urban confound
- Vacancy rate ← supply-side proxy
- Gini index (income inequality)

Standard errors: clustered at the state level (50 clusters).

Diagnostic: are the domains too correlated?

Correlation among ODIS percentile-rank predictors

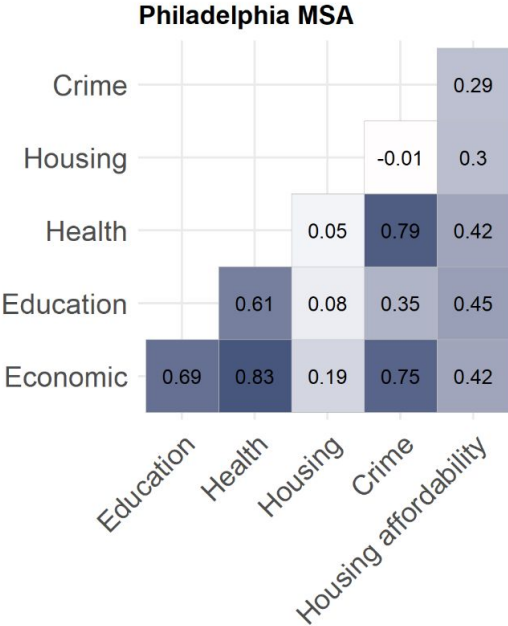
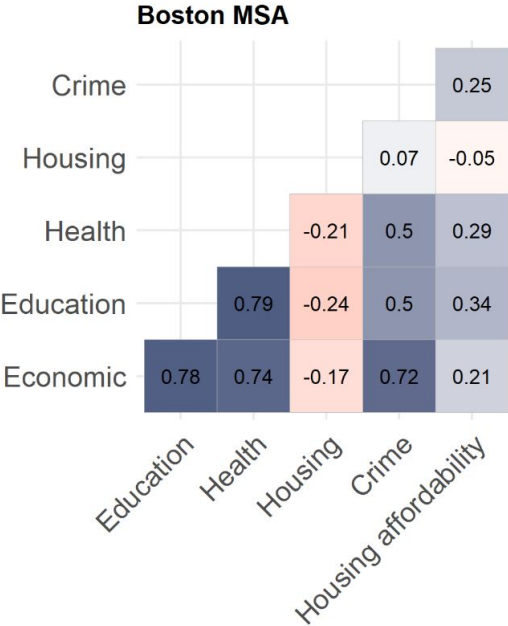
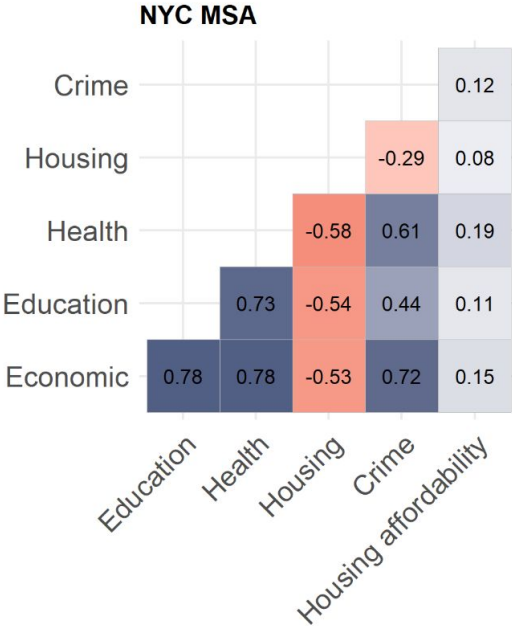


Correlations: 0.40 to 0.63

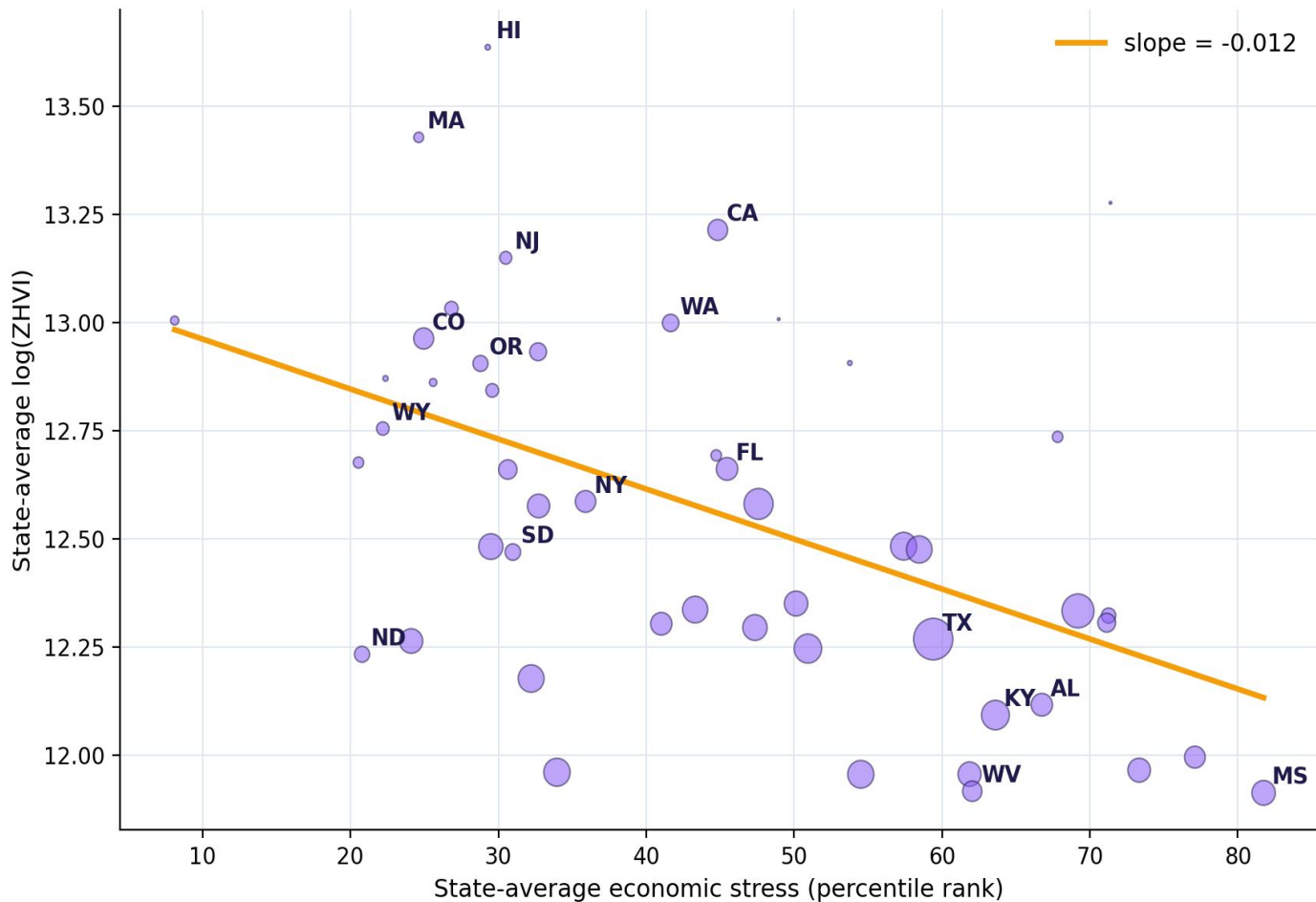
- Moderate, not extreme — VIFs all < 3 in the full model.
- Joint test remains the more reliable inference
- We also run each domain one at a time as a robustness check

Correlation Matrices by MSA

Pairwise correlations: ODIS domains vs. Housing Affordability

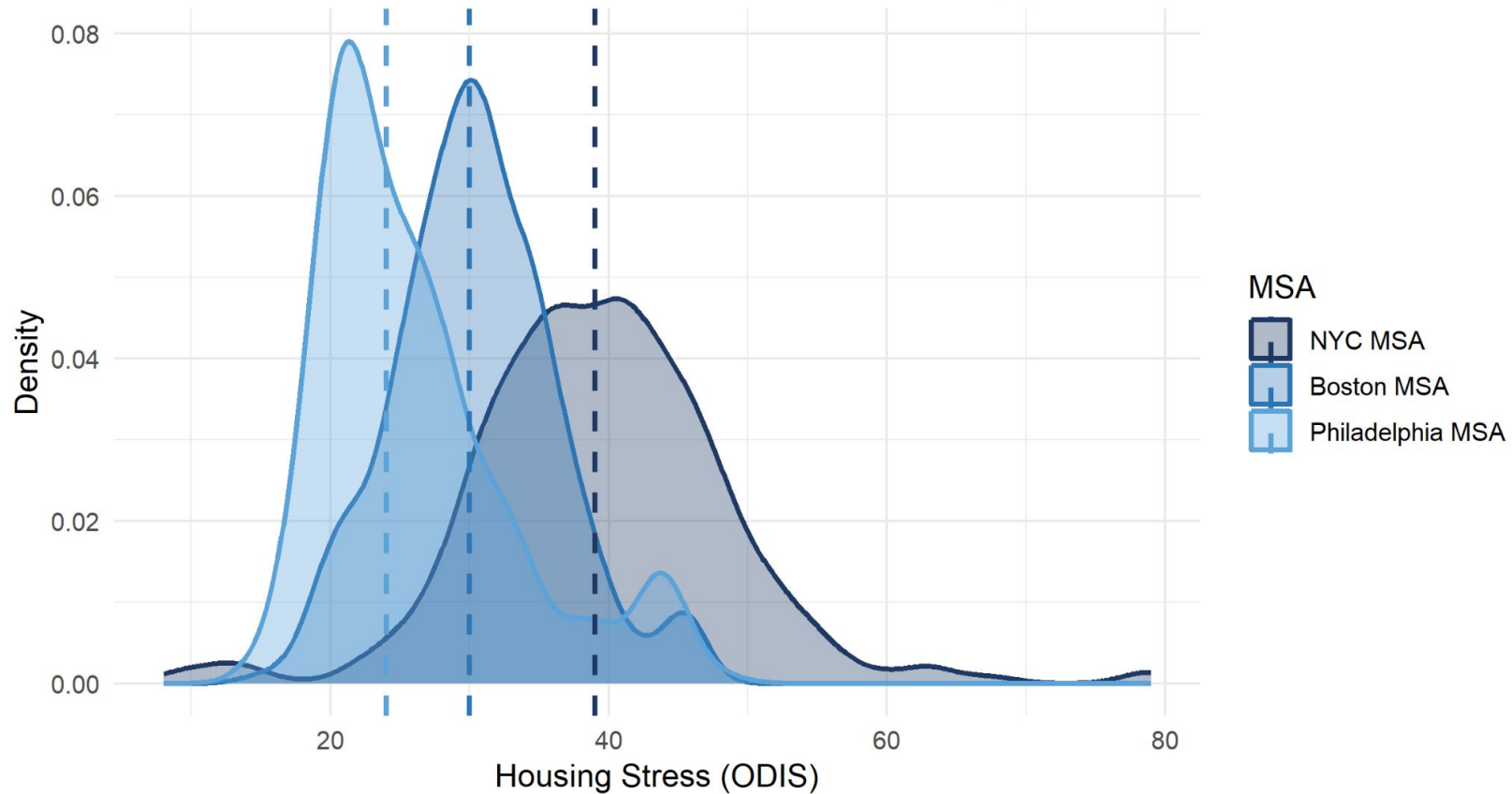


Higher state-average economic stress → lower state-average home values



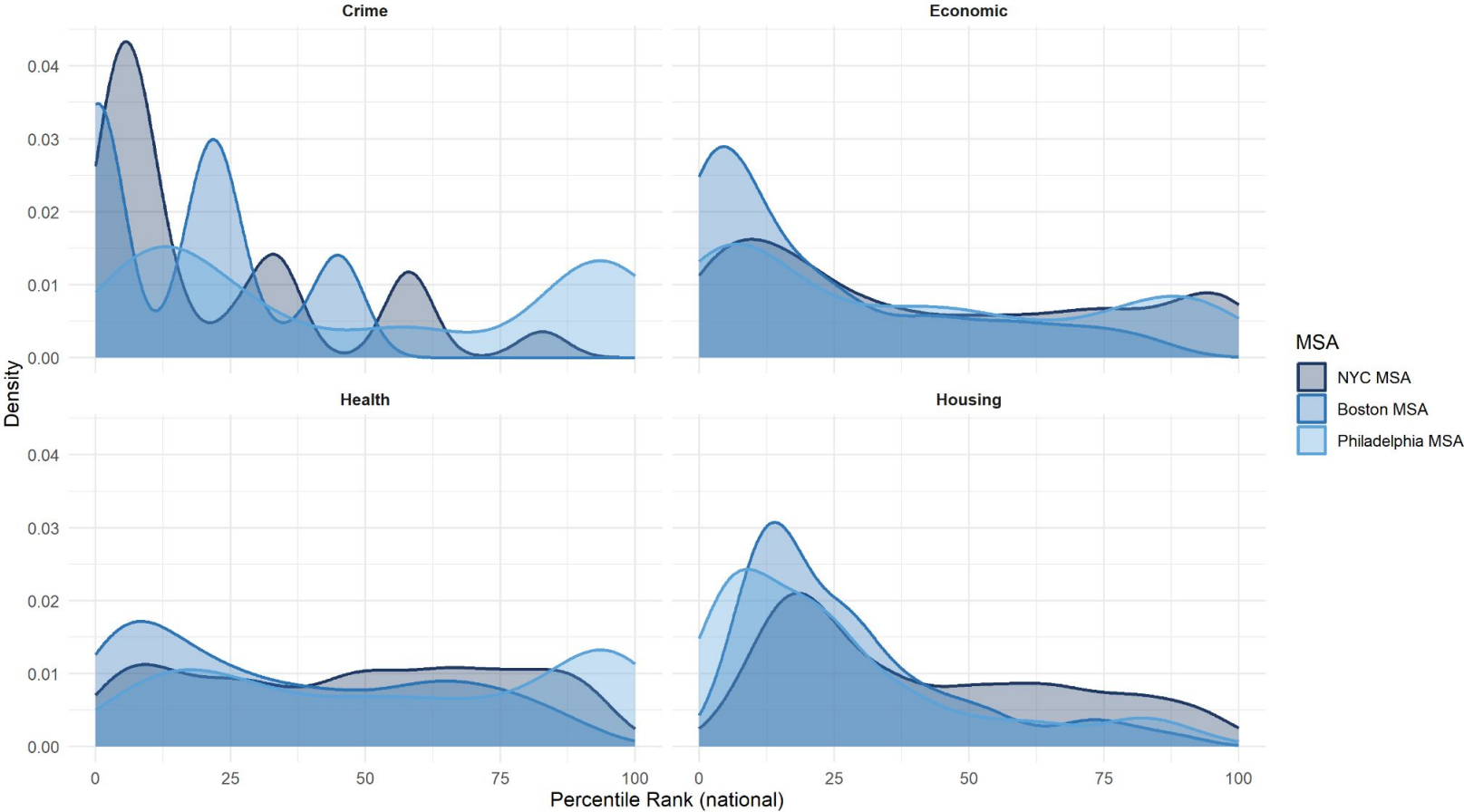
Distribution of Housing Affordability Stress by MSA

Score = % of households spending $\geq 30\%$ of income on housing | Dashed lines = MSA medians



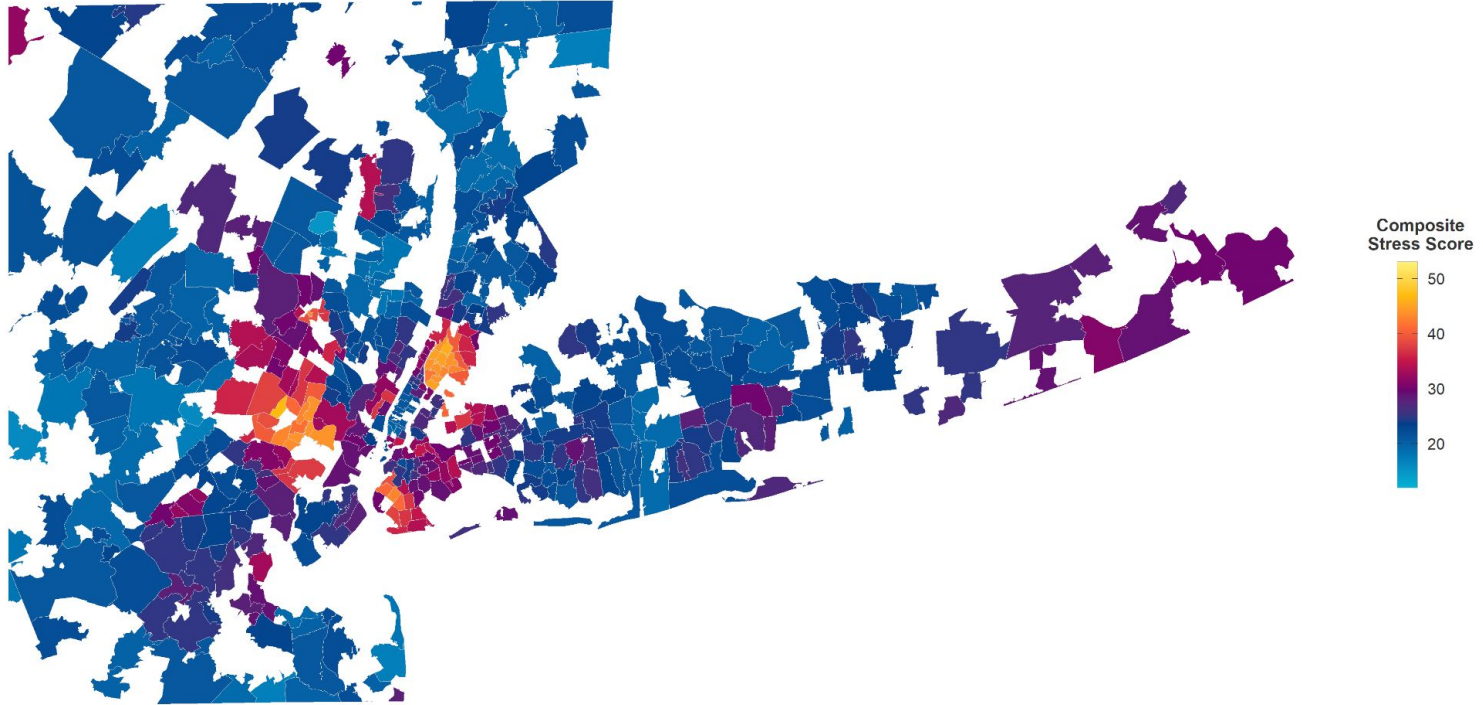
Source: ODIS v2 2024

Percentile Rank Distributions by Domain and MSA
Percentiles relative to all ~23,600 U.S. school neighborhoods in ODIS



New York – Newark – Jersey City MSA

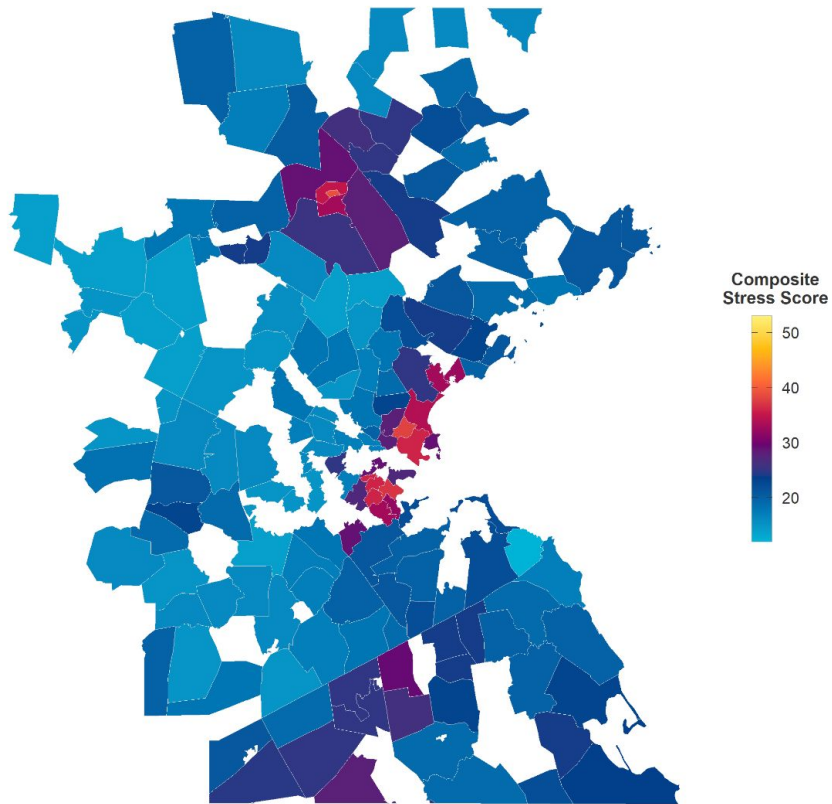
ODIS Composite Stress Score by ZIP Code | Higher = more stress



Source: ODIS v2 2024 (NYU / NAF)

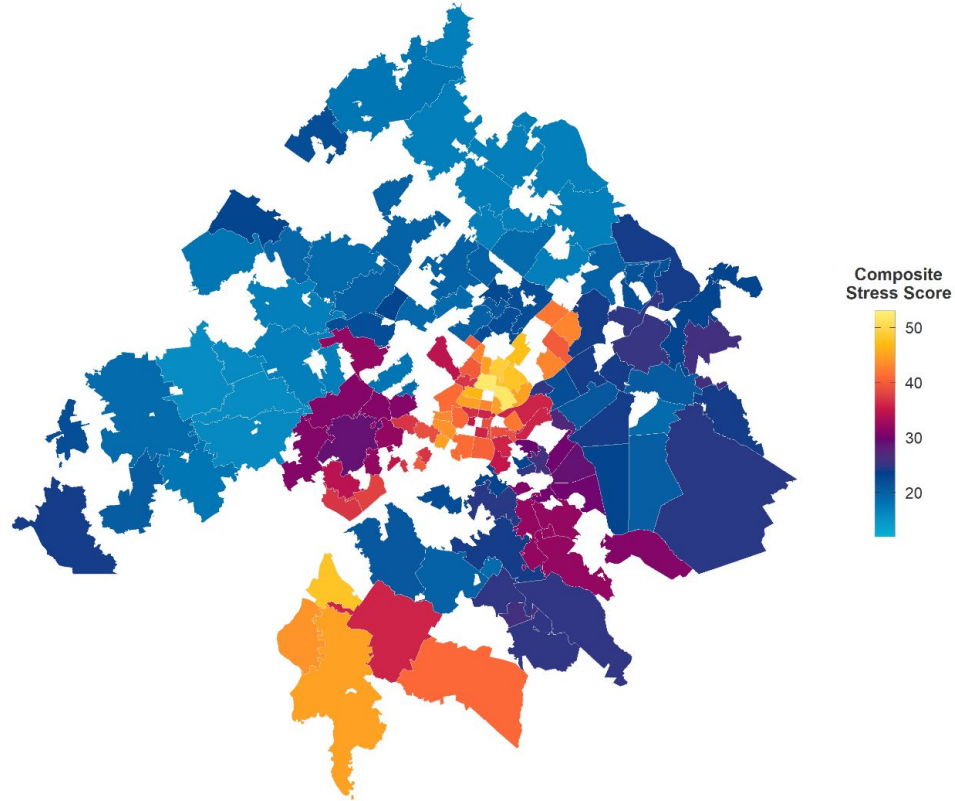
Boston – Cambridge – Newton MSA

ODIS Composite Stress Score by ZIP Code | Higher = more stress

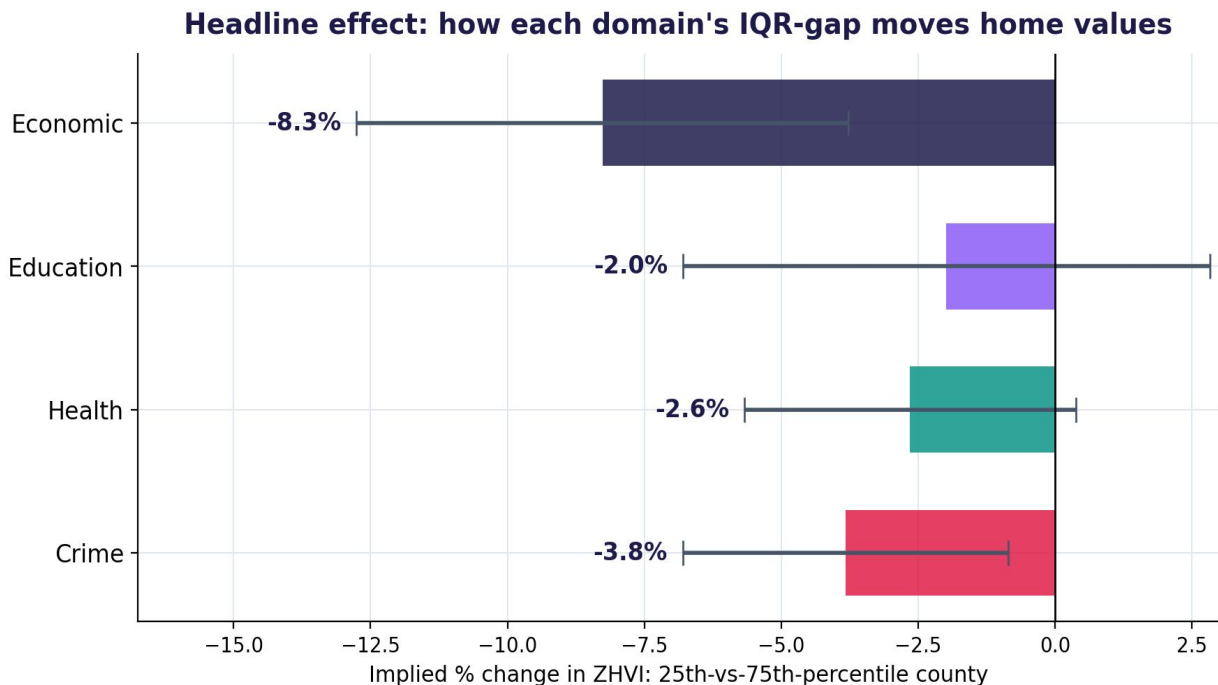


Philadelphia – Camden – Wilmington MSA

ODIS Composite Stress Score by ZIP Code | Higher = more stress



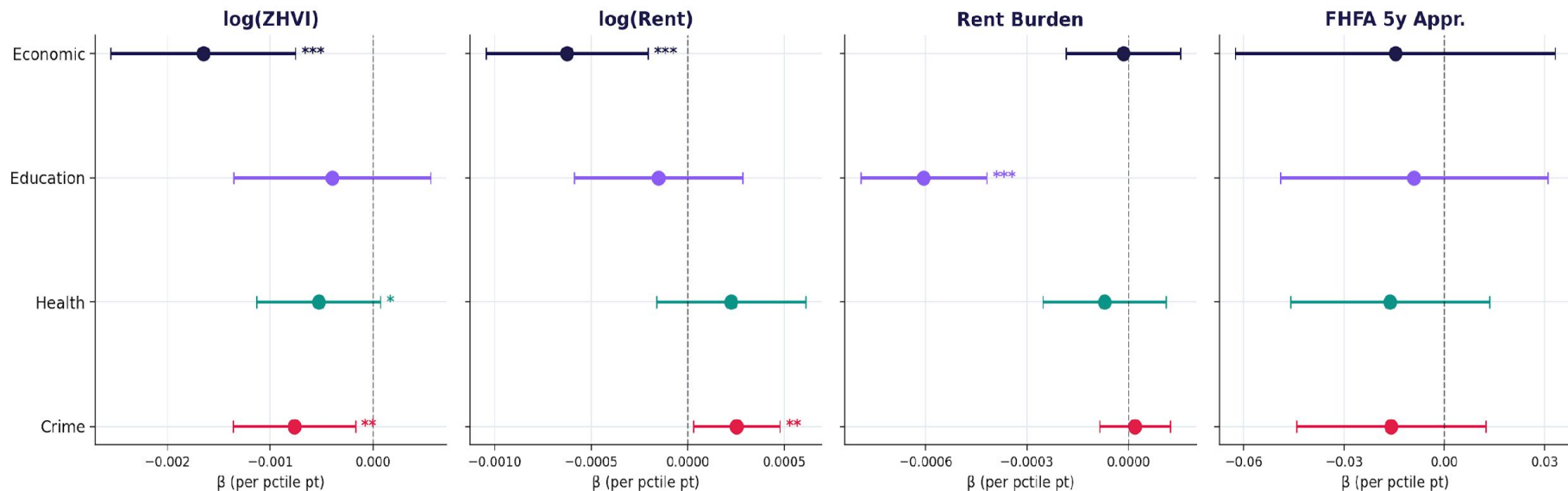
Results: economic stress dominates



A 25th-vs-75th percentile gap in economic stress is associated with ~ 8% lower home values, after controlling for state, income, density, vacancy, and inequality.

Across all four outcomes — different stories

ODIS percentile-rank coefficients across four housing outcomes



Bars = 95% CI; *** p<0.01, ** p<0.05, * p<0.10. State FE + controls.

PRICE LEVELS

Stressors price in (ZHVI, rent)

AFFORDABILITY

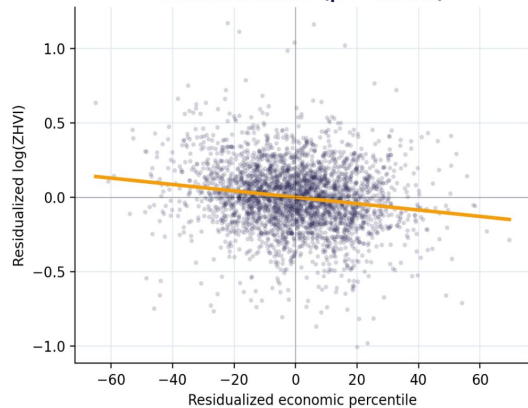
Rent burden — only education stress matters

DYNAMICS

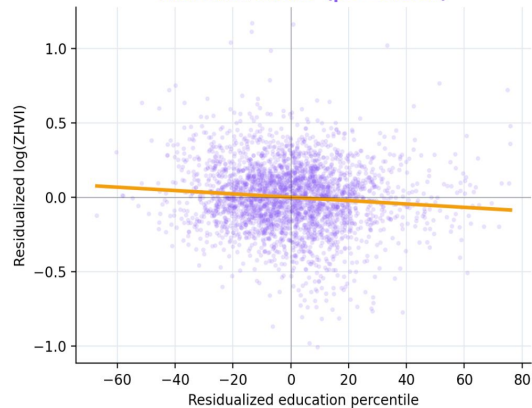
Appreciation: ODIS doesn't predict it (null)

Frisch-Waugh: ODIS domain vs log(ZHVI), partialling out state FE + controls

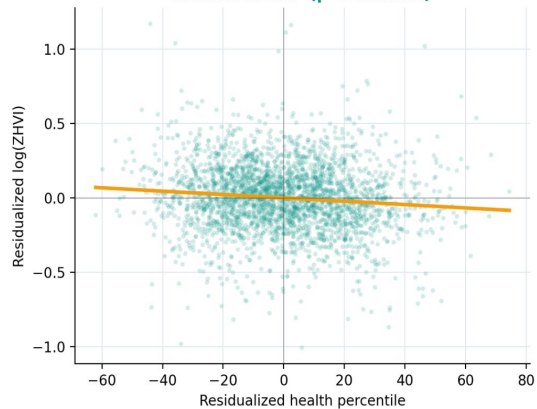
Economic stress ($\beta = -0.0021$)



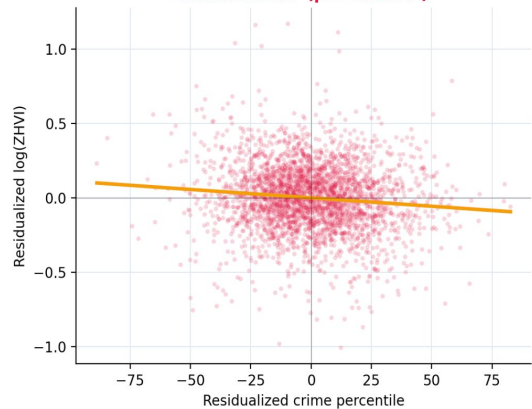
Education stress ($\beta = -0.0011$)



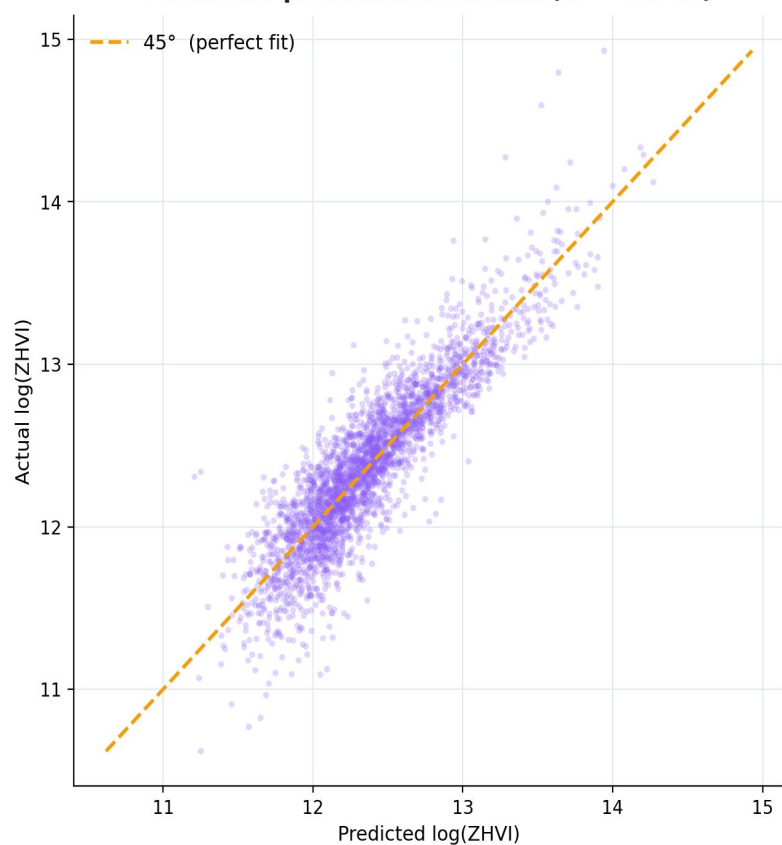
Health stress ($\beta = -0.0011$)



Crime stress ($\beta = -0.0011$)

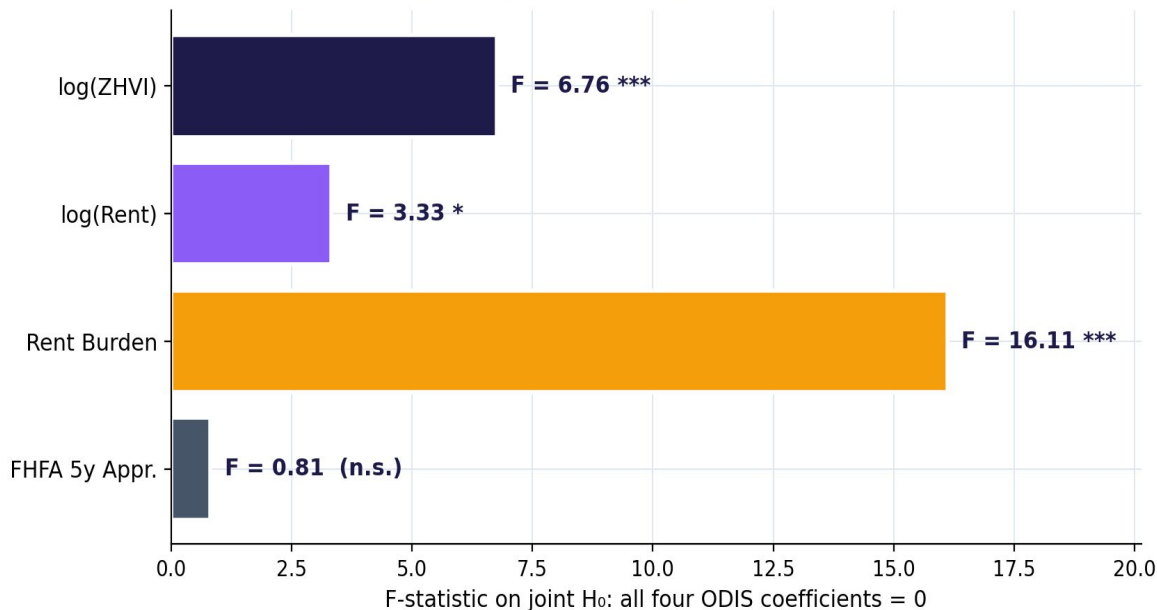


Model fit: predicted vs actual ($R^2 = 0.799$)



Joint significance: H_0 rejected for 3 of 4 outcomes

Joint significance across outcomes



What the F-tests say

- Rent Burden has the strongest joint signal ($F = 16.1$, $p < 1e-8$) even though only education is individually significant.
- ZHVI: $F = 6.76$, $p = 0.0002$.
- Rent levels: $F = 3.33$, $p = 0.02$.
- Appreciation: $F = 0.81$, $p = 0.52$
→ fail to reject. ODIS doesn't predict 5-year appreciation.

Robustness (DV = log ZHVI)

Domain	Solo coef (per 25 pctile)	Solo p-value	Full-model coef (per 25 pctile)	Full model p-value
Economic	-0.0022 ($\approx$ -5.4%)	<0.001	-0.0017 ($\approx$ -4.1%)	<0.001 (stable)
Education	-0.0011 ($\approx$ -2.8%)	0.024	-0.0004 ($\approx$ -1.0%)	0.42 (not stable)
Health	-0.0011 ($\approx$ -2.8%)	0.002	-0.0005 ($\approx$ -1.3%)	0.086 (Partly absorbed)
Crime	-0.0011 ($\approx$ -2.8%)	<0.001	-0.0008 ($\approx$ -1.9%)	0.086 (Partly absorbed)

Why the joint test matters

Each domain alone is strongly negative on log(ZHVI). When all four enter together, education and health get absorbed by economic and crime — they explain the same variation. This is the textbook multicollinearity story: individual coefficients aren't reliable, but the joint hypothesis is.

Table 1: ODIS stress scores on housing outcomes (controls + state FE)

	<i>Dependent variable:</i>			
	log(ZHVI)	log(Rent)	Rent Burden	FHFA 5y Appr.
Economic stress (percentile)	−0.002*** (0.0005)	−0.0006*** (0.0002)	−0.00001 (0.00009)	−0.015 (0.024)
Education stress (percentile)	−0.0004 (0.0005)	−0.0002 (0.0002)	−0.0006*** (0.00010)	−0.009 (0.020)
Health stress (percentile)	−0.0005* (0.0003)	0.0002 (0.0002)	−0.00007 (0.00009)	−0.016 (0.015)
Crime stress (percentile)	−0.0008** (0.0003)	0.0003** (0.0001)	0.00002 (0.00005)	−0.016 (0.014)
log(Population)	0.016 (0.014)	0.043*** (0.010)	0.026*** (0.004)	−0.476 (0.719)
log(Median HH income)	1.126*** (0.082)	0.553*** (0.042)	−0.109*** (0.016)	−5.457*** (2.651)
Gini index	1.117*** (0.312)	0.266** (0.120)	0.096 (0.059)	−41.462*** (9.662)
log(Population density)	0.062*** (0.016)	0.045*** (0.008)	0.018*** (0.004)	0.400 (0.666)
Vacancy rate	0.700*** (0.122)	0.194*** (0.058)	−0.021 (0.020)	28.346*** (4.116)
Constant	−0.742 (0.994)	0.184 (0.445)	1.340*** (0.178)	123.402*** (32.645)
State FE	Yes	Yes	Yes	Yes
Observations	2,913	2,911	2,913	2,590
R ²	0.799	0.841	0.515	0.438
Adjusted R ²	0.795	0.838	0.505	0.425
F Statistic	472.25***	219.00***	72.85***	53.59***
<i>ODIS joint F (4 vars)</i>	6.76***	3.33**	16.11***	0.81

Note:

Cluster-robust SEs (state) in parentheses; *p<0.1; **p<0.05; ***p<0.01

Limitations

Cross-sectional snapshot	We can only estimate associations, not causal effects. No within-county dynamics.
Aggregation & attenuation bias	Schools → counties via the median. True within-neighborhood effects are likely larger; our coefficients are lower bounds.
Multicollinearity in domains	ODIS domains correlate 0.27–0.51 — individual coefficients are noisy, joint tests are reliable.
Crime data coverage	FBI UCR drops ~7% of small/rural counties. Crime score top-coded at 95th percentile.
Health domain heterogeneity	Lead-risk and park-access only cover 500 largest cities — Health score built from different components in rural vs urban counties.
Omitted variables	Housing-supply elasticity, climate exposure, redlining history, racial composition. State FE absorbs only the state mean.

TAKEAWAYS

Neighborhood stressors capitalize meaningfully into home-value levels — economic stress dominates ($\approx 8\%$ IQR gap on ZHVI).

Rent levels and rent burden tell different stories: stressors price in less cleanly, and affordability is shaped largely by income, not stressors.

ODIS does not predict 5-year appreciation — short-run dynamics are driven by income / market cycles, not neighborhood quality.

Joint tests are the right inference: domain coefficients individually are noisy because the four are correlated, but together they're highly significant for 3 of 4 outcomes.

Future work: ODIS V1 (2022) $\times$ V2 (2024) panel $\rightarrow$ county fixed effects $\rightarrow$ causal identification.